

Landscape Analysis of Physical Modeling of Eco-Geological Sounds

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1 Narrative

When combining our bibliographies, we realized that our examination was leaning towards the synthesis of ecological in addition to geological sounds. Because we feel that these two areas of study are intrinsically linked, we decided to broaden our research to include both, allowing us another degree of freedom in the exploration of sources. Thus, we arrived at the term eco-geological.

We chose to divide the sources according to the sequential stages of our research plan. Each category is a step of the process, with Category 1 helping us familiarize with what tools for physical modeling exist, Category 2 giving a framework on how those tools have been applied in other projects we could emulate and Category 3 providing areas our research could extend into in terms of eco-geological exploration, even outside of strictly sound synthesis.

Category 1 examines the development tools that will serve as the foundation of our research. While some like C++ are familiar, others present a gap in our knowledge. Exploring these tools will provide us an opportunity to improve our understanding of digital signal processing (DSP) and enhance our technical and programming skills. [5] An understanding of the works in Category 2 will function as a baseline, allowing us to apply the skills acquired in Category 1 in preparation for exploring the contents of Category 3. For example, understanding a methodology for synthesizing object collisions could prove useful when trying to replicate the sound made when two rocks are rubbed together in a geological context. [9] When seeking applications of physical modeling synthesis specifically in the realm of geological sounds, we realized that the topic was somewhat underexplored, as examples of other synthesis types, such as granular synthesis, are more prevalent. [4] Works that reference describe the targeted sounds as "environmental" are also more numerous, as opposed to the somewhat specific "geological" descriptor. [6]

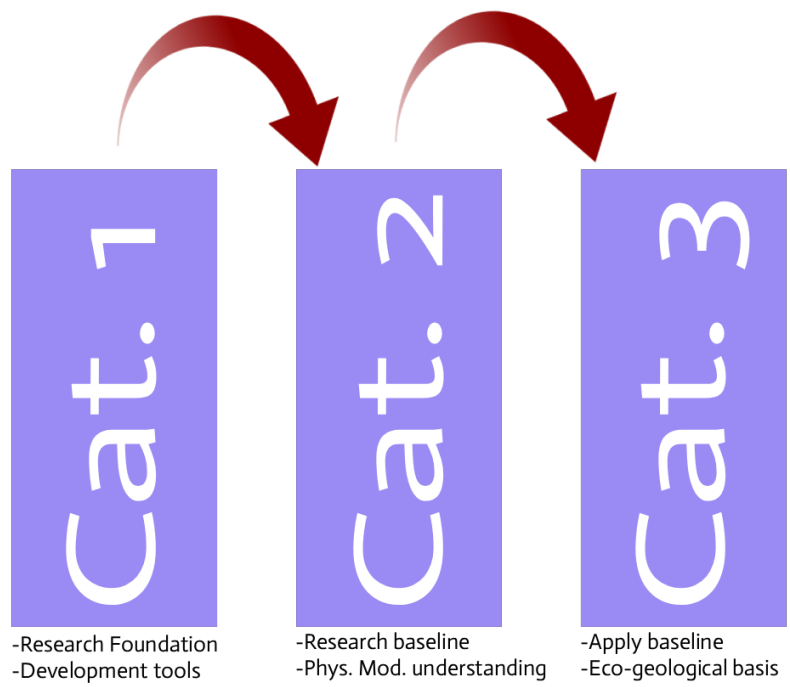


Figure 1: Visual aid showing linear approach.

2 Category 1

2.1 The Synthesis ToolKit in C++

STK is a long-running open-source C++ class library (Princeton/McGill) providing physical-modeling instrument classes (plucked/bowed strings, winds, brass, percussion) alongside general DSP unit generators, real-time audio/MIDI I/O, and file output. Its lasting contribution is portability and reuse: STK’s instrument classes have been embedded in numerous other environments (Chuck, Csound, Max/MSP, SuperCollider, Faust), making it one of the most widely propagated physical-modeling codebases in computer music.

2.2 Physical modeling for pure data (PMPD) and real time interaction with an audio synthesis

This paper introduces PMPD, a Pure Data library that simulates mass-spring particle networks (Newtonian “mass” and visco-elastic “link” objects) as a control-rate physical model, rather than an audio-rate synthesis engine. Its contribution is treating the physical simulation as a flexible, intuitive black box that a performer can touch, distort, and drive in real time, with its motion then mapped onto an external audio synthesis engine (e.g., via additive synthesis or scanned synthesis). [3]

2.3 Synth-A-Modeler compiler: Modular and open-source sound synthesis using physical models

An open-source compiler that lets artists build physical-modeling DSP instruments from analog-style specifications rather than signal graphs. [1]

2.4 The Faust physical modeling library: A modular playground for the digital luthier.

This paper introduces Faust’s `physmodels` library, giving composers/programmers modular building blocks (waveguides, mass-spring elements, string/wind components) to assemble physical models of existing or entirely novel instruments in the Faust language. It also demonstrates combining this library with `mesh2faust` — a tool that generates physical models directly from 3D mesh data — to build a marimba model, linking geometric/CAD design to sound synthesis. [5]

3 Category 2

3.1 Foley Automatic

Algorithms for real-time synthesis of realistic contact sound effects (impact, sliding, rolling) for animation and games, driven automatically by a 3D dynamics simulation and user interaction. [10]

3.2 Synthesis of shape dependent sounds with physical modeling

This paper simulates the sounds of colliding rigid objects (chairs, tables, rods) using a linear modal model: an object’s frequency spectrum, eigenmodes, and a single internal-friction parameter fully determine its impact sound as a sum of exponentially decaying sinusoids. Its key contribution is making the synthesized sound depend on *where* the object is struck, not just its material and shape, and demonstrating this in a real-time interactive “Sonic Explorer” environment. [9]

3.3 Physical modeling using digital waveguides

This is the foundational paper establishing digital waveguide synthesis: representing a vibrating string, bore, or tube as pairs of delay lines carrying left- and right-traveling samples of the wave equation’s D’Alembert solution. Its key contribution is showing that physically exact (lossless, linear) wave propagation can be simulated with extremely low computational cost, making real-time physical modeling of instruments practical — this became the basis for products like the Yamaha VL1. [8]

3.4 Physics-Based Crumpling Sounds Model

A physics-based model of crushing, walking, and running sounds built up from the sound of a single crumpling event. Includes the Pd module used to implement the crushing-can model. [2]

4 Category 3

4.1 Physical Modeling of Environmental Sounds

Automatically generating everyday environmental sounds from physical models rather than recordings using impact sounds (spherical objects striking flat plates or beams) as its main case study and comparing the synthesized spectra against real recordings. [6]

4.2 Ecologically-based granular synthesis.

This paper proposes a granular synthesis technique that builds environmental sounds (bouncing, water streams, scraping) from short sampled grains driven by “ecologically” constrained control functions rather than purely deterministic or stochastic ones. [4]

4.3 Sounds of the Earth’s Crust

MIT geologists crush marble samples and record the sounds they emit to fingerprint rock stability with depth. Not really sound synthesis, but a cool example

of how physics maps onto sound. [7]

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